

Simulation Assumption Basis Module (SABM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Assumption Basis

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Governed Space

Simulation Assumption Basis

Canonical Definition

Simulation Assumption Basis Module (SABM) defines the governance framework for establishing, documenting, qualifying, and maintaining the basis upon which an identified Simulation Assumption is introduced, accepted, retained, modified, or rejected within a simulation.

It establishes the methodological mechanism through which a Simulation Assumption is connected to the rationale, observations, knowledge, evidence references, models, prior experience, expert judgment, theoretical foundations, engineering constraints, or other grounds that explain why the assumption is being used.

SABM does not determine whether an assumption is universally true.

It governs whether the assumption has a sufficiently explicit and traceable Simulation Assumption Basis for its defined role inside the simulation.

Operational Role

SABM serves as the second internal governance space of the Simulation Assumption Standard (SAS).

The preceding SAIM — Simulation Assumption Identification Module establishes:

Which assumptions exist and which are materially relevant to the simulation?

SABM asks:

On what basis is each identified Simulation Assumption being used, and is that basis sufficiently explicit, relevant, bounded, and traceable for its role within the simulation?

The governed progression is:

Identified Simulation Assumption → Basis Discovery → Basis Attribution → Basis Classification → Basis Relevance → Basis Sufficiency → Basis Limitation → Assumption-Basis Record

The resulting governed basis supports subsequent examination of:

Assumption Dependency → Assumption Uncertainty → Assumption Limitation

Module Operational Space

SABM governs: assumption basis identification, assumption rationale, basis attribution, basis origin, basis classification, empirical basis, theoretical basis, historical basis, observational basis, engineering basis, expert-judgment basis, model-derived basis, inherited basis, precedent-based basis, constraint-derived basis, mixed basis, basis relevance, basis applicability, basis sufficiency, basis currency, basis specificity, basis strength representation, unsupported assumptions, weakly supported assumptions, conditionally supported assumptions, conflicting assumption bases, basis limitations, basis change, basis-to-assumption traceability, and maintenance of the Simulation Assumption Basis Record.

Module Function

SAIM makes assumptions visible.

SABM makes the reason for using them visible.

A simulation may explicitly document an assumption while providing no explanation for why that assumption was selected.

For example:

Battery performance is assumed to remain constant throughout the simulated mission.

The statement identifies an assumption.

It does not establish: why constant performance was selected, which operating conditions support it, whether historical observations support it, whether it

derives from engineering specifications, whether it is merely a modeling convenience, or whether contrary information exists.

SABM prevents the presence of an assumption statement from being mistaken for the existence of an assumption basis.

Simulation Assumption Basis

A Simulation Assumption Basis is the governed rationale or supporting foundation explaining why a Simulation Assumption is introduced or retained for a defined simulation context.

The basis may contain one or several supporting elements.

Conceptually:

Simulation Assumption

↓

Why is this premise being used?

↓

Simulation Assumption Basis Basis Does Not Equal Truth

SIMULOS® establishes:

A documented basis explains why an assumption is used; it does not automatically establish that the assumption is true.

An assumption may possess a rational basis while still containing substantial uncertainty.

This distinction preserves the boundary between SABM and later assumption governance.

Basis Types

A Simulation Assumption may derive from different basis types.

Empirical Basis

Derived from observations, measurements, experiments, operational records, or other empirical information.

Theoretical Basis

Derived from an accepted or proposed theoretical model, scientific principle, mathematical relationship, or conceptual framework.

Historical Basis

Derived from historical behavior, prior events, previous simulations, or longitudinal observations.

Engineering Basis

Derived from technical specifications, design properties, engineering constraints, tolerances, or established engineering reasoning.

Expert-Judgment Basis

Derived materially from the judgment of one or more qualified persons.

Model-Derived Basis

Derived from another model, simulation, analytical system, or computational representation.

Inherited Basis

Transferred from a prior simulation, external component, reused simulation asset, or existing model.

Precedent-Based Basis

Derived from previous accepted practice, comparable cases, or established operational precedent.

Constraint-Derived Basis

Introduced because available resources, data, model capabilities, computation, time, or simulation design make a more complete representation impractical.

Mixed Basis

Derived from several materially relevant basis types.

Basis Attribution

Each material assumption should be attributable to the basis supporting it.

The minimum relationship is:

Assumption → Basis

Where several basis elements exist:

Assumption → Basis Element A + Basis Element B + Basis Element C

Where several assumptions depend upon one basis:

Basis → Assumption A + Assumption B + Assumption C

This allows later changes in the basis to propagate to affected assumptions.

Basis Origin

Where available, SABM identifies where the basis originated.

Possible origins include: measurement systems, operational records, scientific literature, engineering documentation, subject-matter experts, external models, prior simulations, organizational experience, technical standards, vendor information, environmental observations, AI-generated analysis, or internal simulation design decisions.

Origin identification improves traceability without automatically establishing reliability.

Basis Relevance

A basis must be relevant to the specific assumption it supports.

For example:

historical performance under normal temperature conditions may provide some basis for an assumption about normal-temperature operation.

It may provide little basis for the same assumption under extreme heat.

SABM therefore asks:

Does the identified basis actually address the condition represented by the assumption?

Basis Applicability

A basis may be valid only within a bounded context.

Relevant applicability dimensions may include: object version, model version, population, environment, operating range, geography, temporal period, scenario, configuration, system state, physical conditions, or behavioral context.

A basis should not silently expand beyond the context from which it was derived.

Basis Sufficiency

SABM evaluates whether the identified basis is sufficiently developed to justify using the assumption for its intended simulation role.

Basis sufficiency is contextual.

An assumption used for exploratory simulation may legitimately operate on a weaker basis than an assumption controlling a high-consequence simulation conclusion.

SABM therefore evaluates sufficiency relative to:

Simulation Purpose + Simulation Scope + Assumption Materiality + Consequence
Basis Strength Representation

Where useful, a Simulation Assumption Basis may be represented as:

Strongly Established Basis

Sufficient Basis

Conditional Basis

Limited Basis

Weak Basis

Unsupported

or:

Basis Undetermined

These states describe the governance condition of the assumption basis.

They do not constitute universal validation of the assumption itself.

Unsupported Assumption

An Unsupported Simulation Assumption is an identified assumption for which no sufficiently identifiable basis has been established.

Unsupported does not automatically mean false.

It means:

the simulation depends upon a premise whose supporting basis has not been sufficiently established.

This condition must remain visible.

Weakly Supported Assumption

A Weakly Supported Simulation Assumption possesses an identifiable basis, but that basis may be: indirect, incomplete, outdated, contextually weak, narrow, dependent upon judgment, poorly traceable, or insufficient relative to the consequence of the assumption.

Weak support should not be represented as strong support merely because the simulation requires the assumption to proceed.

Conditionally Supported Assumption

An assumption may have sufficient basis only while specified conditions remain true.

For example:

Sensor degradation remains negligible during the simulated mission

may be supported only where:

temperature remains within the qualified operating range.

The assumption basis should therefore preserve the relevant condition.

Basis Currency

A basis may lose relevance over time.

Examples include: outdated operational records, obsolete engineering specifications, changed environments, changed system versions, updated scientific knowledge, changed populations, or superseded models.

SABM therefore preserves the temporal relationship between the basis and the simulation.

Inherited Basis Governance

An assumption inherited from another simulation may arrive with an inherited basis.

SABM does not automatically accept that basis.

It asks whether the inherited basis remains applicable to: the current Simulation Object, current purpose, current scope, current representation, current model, current environment, and current simulation conditions.

Expert-Judgment Basis

Expert judgment may constitute a legitimate Simulation Assumption Basis.

Where material, SABM should make visible: the domain of expertise, the question addressed, the context of the judgment, material conditions, known limitations, and whether materially different expert judgments exist.

The existence of expertise should not transform judgment into empirical fact.

Model-Derived Basis

A simulation assumption may be supported by another model.

This creates a methodological chain:

External Model → Derived Basis → Simulation Assumption → Current Simulation

Where the external model changes materially, assumptions dependent upon that model may require reassessment.

Constraint-Derived Assumptions

Some assumptions exist because the simulation cannot practically represent everything.

For example:

a computationally expensive physical interaction may be simplified into a stable approximation.

The basis may therefore be:

Simulation Constraint + Acceptable Approximation Rationale

SABM requires this rationale to remain visible rather than presenting the approximation as established reality.

Convenience Is Not Sufficient Basis by Itself

SIMULOS® establishes:

An assumption should not be represented as methodologically supported merely because it makes the simulation easier to construct or execute.

Modeling convenience may explain why an assumption exists.

It does not automatically establish that the assumption is appropriate.

Authority Is Not Sufficient Basis by Itself

An assumption should not receive unlimited methodological strength solely because it originates from: a senior person, a recognized institution, a regulator, a vendor, a previous project, an established model, or a widely used practice.

Authority may contribute to the basis.

The assumption remains bounded by the relevance and applicability of that basis.

Repeated Use Is Not Validation

SIMULOS® establishes:

Repeated use of an assumption does not transform the assumption into an established fact.

An assumption used across many simulation generations may still require reassessment when: the object changes, reality changes, the environment changes, new evidence emerges, or the intended simulation use changes.

Conflicting Assumption Basis

Different basis elements may support materially different assumptions.

For example:

historical observations may suggest one behavioral range,

while:

recent operational measurements suggest another.

SABM requires material conflict to remain visible.

A conflict should not be silently resolved by selecting the basis that produces the preferred simulation outcome.

Basis Conflict Record

Where material conflict exists, SABM may preserve:

Assumption

Basis A

Basis B

Nature of Conflict

Materiality

Current Treatment

and:

Required Reassessment Basis Limitation

Every assumption basis may contain limitations.

Examples include: narrow sample, incomplete observations, environmental mismatch, outdated information, uncertain extrapolation, limited expertise, model dependency, incomplete traceability, or reliance upon indirect evidence.

SABM identifies these basis limitations.

The broader consequences of the assumption itself are governed subsequently by SALM — Simulation Assumption Limitation Module.

Basis-to-Assumption Traceability

SABM should preserve:

Basis Origin

↓

Basis Element

↓

Applicable Context

↓

Simulation Assumption

↓

Affected Simulation Element

This allows later reviewers to reconstruct why the assumption entered the simulation.

Basis Change

A basis may materially change.

Examples include: new measurements, new scientific findings, revised engineering specifications, changed expert judgment, corrected historical data, new operational experience, or replacement of an external model.

Material basis change should trigger reassessment of affected assumptions.

Basis Change Propagation

The governance relationship is:

Basis Change → Affected Assumption → Affected Simulation Element → Reassessment Need

SABM identifies the propagation path.

It does not itself govern the complete resimulation lifecycle.

Basis Expiration

Some assumption bases may have an explicit or implicit useful period.

Where a basis is time-sensitive, its continued use should be reassessed when: the defined period expires, the underlying reality materially changes, or the

Simulation Object materially changes.

Basis Rejection

A previously accepted assumption basis may become unsuitable.

Possible causes include: contradiction by stronger information, invalid source, changed context, model replacement, discovered methodological defect, or loss of applicability.

Basis rejection should not erase historical use.

The prior basis should remain traceable to simulations that relied upon it.

Assumption Basis Record

For every material Simulation Assumption, SABM may establish a Simulation Assumption Basis Record containing: Assumption ID, Basis ID, Basis Statement, Basis Type, Basis Origin, Applicable Context, Supporting References, Basis Relevance, Basis Sufficiency, Basis Currency, Basis Conditions, Basis Limitations, Basis Conflicts, Basis Status, and traceability references.

Assumption Basis Status

A basis may carry a status such as:

Identified

Under Assessment

Sufficient

Conditional

Limited

Weak

Unsupported

Conflicted

Superseded

Rejected

or:

Undetermined Basis Sufficiency Gap

A Simulation Assumption Basis Gap exists where: no basis is identifiable, the basis does not address the assumption, the basis is materially outdated, applicability cannot be established, material conflict remains unresolved, or the basis is insufficient relative to the role of the assumption.

The gap must remain visible to downstream simulation governance.

No Basis Laundering Principle

SIMULOS® establishes:

A weak, indirect, inherited, conditional, or disputed basis must not become represented as strong merely because it passes through a simulation model.

The simulation does not strengthen the basis simply by consuming it.

No Circular Basis Principle

An assumption should not be represented as supported where its basis ultimately depends upon the same assumption.

For example:

Assumption A is accepted because Model X predicts it.

But:

Model X predicts it because Model X itself assumes A.

This creates circular support.

SABM requires material circularity to remain visible.

No Output-as-Basis Circularity

Simulation output should not automatically become the basis proving the assumption that generated that output.

Conceptually:

Assumption → Simulation → Output

cannot, without independent justification, become:

Output → Proof of Assumption

This distinction is essential to simulation governance.

Assumption Basis Sufficiency Assessment

At the completion of SABM, each material assumption should be capable of receiving a basis assessment such as:

Basis Sufficient

Basis Conditionally Sufficient

Basis Limited

Basis Insufficient

Basis Conflicted

or:

Basis Undetermined

This assessment becomes an input to subsequent assumption governance.

Minimum Implementation Framework

1. Bind the Basis Inquiry to the Assumption

Identify the exact Simulation Assumption and simulation context being assessed. 2. Identify the Supporting Basis

Determine why the assumption is being used and identify the material basis elements. 3. Classify and Attribute the Basis

Establish basis type, origin, and relationship to the assumption. 4. Assess Relevance and Applicability

Determine whether the basis actually supports the assumption within the applicable simulation context. 5. Assess Sufficiency

Evaluate whether the basis is sufficiently developed relative to the assumption's materiality, simulation purpose, scope, and consequence. 6. Identify Conditions, Conflicts, and Limitations

Preserve material qualifications rather than converting them into unconditional support. 7. Establish Basis Status

Record whether the basis is sufficient, conditional, limited, insufficient, conflicted, or undetermined. 8. Preserve Basis-to-Assumption Traceability

Maintain the relationship:

Basis → Assumption → Affected Simulation

Element 9. Govern Basis Change

Identify material changes capable of requiring assumption reassessment. 10. Transfer the Governed Basis Forward

Provide the qualified Simulation Assumption Basis to dependency, uncertainty, and limitation governance.

Governance Outputs

SABM may produce: Simulation Assumption Basis Record, Assumption Basis Identifier, Assumption Basis Classification, Assumption Basis Origin Record, Empirical Basis Record, Theoretical Basis Record, Historical Basis Record, Engineering Basis Record, Expert-Judgment Basis Record, Model-Derived Basis Record, Inherited Basis Record, Constraint-Derived Basis Record, Assumption

Basis Relevance Assessment, Assumption Basis Applicability Assessment, Assumption Basis Sufficiency Assessment, Assumption Basis Currency Assessment, Assumption Basis Condition Record, Assumption Basis Limitation Record, Assumption Basis Conflict Record, Assumption Basis Gap Record, Assumption Basis Change Record, Assumption Basis Status, Assumption Basis Traceability Map, and Governed Simulation Assumption Basis Set.

Use Case 1 — Autonomous Robot Simulation

Scenario

A warehouse robot simulation assumes:

Human workers will maintain a minimum reaction distance under normal operating conditions.

The assumption is material because collision-avoidance performance depends upon it. Application

SABM identifies the basis as a combination of: historical facility observations, worker-movement measurements, and an existing behavioral model.

The records show that the measurements were collected during normal operation and do not cover emergency evacuation conditions. Result

The assumption receives:

Basis Conditionally Sufficient — Normal Operating Conditions

The simulation cannot legitimately treat the same basis as supporting emergency-condition behavior without additional grounds. Use Case 2 — Space Mission Simulation

Scenario

A spacecraft simulation assumes:

A communication subsystem will remain available for at least 99% of the simulated mission window. Application

SABM identifies: engineering specifications, historical subsystem performance, environmental qualification data, and mission-design constraints

as the assumption basis.

However, the historical observations do not include the full radiation environment expected during the proposed mission. Result

The basis remains explicit and traceable but is classified:

Limited Basis — Radiation Environment Coverage Incomplete

The limitation can then propagate into subsequent assumption uncertainty and limitation governance.

Architectural Position

Simulation Assumption Basis is the second internal governance space of the Simulation Assumption Standard (SAS).

The complete SAS progression is:

Simulation Assumption Identification → Simulation Assumption Basis →
Simulation Assumption Dependency → Simulation Assumption Uncertainty →
Simulation Assumption Limitation

SAIM establishes:

Which assumptions exist?

SABM establishes:

Why is each assumption being used and what supports it?

SADM establishes:

What depends upon each assumption and how do assumptions interact?

SAUM establishes:

How uncertain is each assumption and how must that uncertainty remain visible?

SALM establishes:

Where does each assumption cease to apply and what limitations follow?

Together, these modules prevent assumptions from functioning as invisible or unsupported premises within simulation design, execution, output interpretation, and downstream reliance.

Foundational Principle

A simulation assumption becomes methodologically supportable only when the basis for using that assumption is explicit, relevant to the simulation context, sufficiently bounded, and traceable to the assumption it supports.

Canonical Closing Statement

Simulation Assumption Basis Module establishes the supporting-basis layer of assumption governance within SIMULOS® by governing why identified Simulation Assumptions are introduced, retained, modified, or rejected within a simulation. Through basis identification, classification, attribution, relevance, applicability, sufficiency, currency, conditions, conflict, limitation, change governance, anti-circularity controls, and complete basis-to-assumption traceability, SABM prevents modeling convenience, inherited practice, authority, repetition, or simulation output from being mistaken for sufficient methodological support and ensures that every material assumption carries a visible and bounded basis into subsequent dependency,

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